

Raymond Gao

(757) 243-6768 | raymond_gao@berkeley.edu | [LinkedIn](#)

EDUCATION

UC Berkeley, College of Computing, Data Science, and Society
Major: Computer Science, BA | Minor: Data Science, BA

Berkeley, CA
May 2029

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++, SQL

Libraries & Frameworks: PyTorch, NumPy, Pandas, Scikit-learn, Matplotlib, ROOT, CUDA

Tools: Git, GitHub, Docker, Visual Studio, IntelliJ

WORK EXPERIENCE

Edwards Lifescience Foundation Cardiovascular Innovation and Research Center

Irvine, CA

Software Engineering Intern

June 2026 - August 2026

- Unified two legacy C++ imaging systems into a single multimodal platform, eliminating duplicate codebases and enabling shared architecture for real-time cardiovascular imaging.
- Integrated CUDA-based signal reconstruction and real-time visualization into the multimodal imaging pipeline, enabling GPU-accelerated conversion of raw acquisition data into live image displays.
- Developed an event-driven coordination layer for a four-thread C++ acquisition pipeline, implementing synchronized startup, cooperative shutdown, and asynchronous error propagation to eliminate shutdown hangs and GPU resource leaks.

Lawrence Berkeley National Laboratory

Berkeley, CA

Simulation & Modeling Engineering Intern

November 2025 - May 2026

- Built C++/ROOT data processing and analysis pipelines to process and evaluate 25,000+ simulated particle collision events for detector performance validation.
- Automated matching pipeline between reconstructed and simulated data, enabling reproducible validation across 50+ datasets.
- Implemented visualization and validation tools to generate performance metrics and evaluate the impact of detector geometry and vibration effects on tracking accuracy.

Thomas Jefferson National Accelerator Facility

Newport News, VA

Software Engineering Intern

October 2024 - May 2025

- Engineered Python ETL pipelines for automated processing and validation of high-volume collider simulation data.
- Optimized petabyte-scale datasets by building automated filtering and visualization workflows to identify key patterns in high-dimensional collider event data.
- Designed a modular evaluation framework for reconstruction accuracy across varying Q^2 levels, enabling reusable validation and visualization workflows.

Dartmouth Hitchcock Medical Center

Lebanon, NH

Machine Learning Intern

June 2024 - August 2024

- Trained and evaluated PyTorch classification models to distinguish typical vs. atypical squamous cells, implementing class-weighted optimization to address class imbalance and achieving 98.5% ROC-AUC on the validation set.
- Built preprocessing workflows in Pandas and NumPy to clean and organize 35+ pathology images for deep learning classification.

CODING PROJECTS

N-Gram Map & Wordnet | Java, IntelliJ, Git

March 2026

- Engineered a time-series analytics system for querying and visualizing historical trends of 40,000+ English words.
- Designed and implemented graph and map-based data structures to efficiently store and process large text datasets.
- Developed backend queries with constant time complexity and integrated them with a web application to support interactive word relationship and trend exploration for user interactions.

Build Your Own World | Java, IntelliJ, Git

April 2026

- Built a game that randomly generates distinct 2D worlds from a user-provided seed.
- Designed a multi-floor elevator system for exploration and toggleable map views to enhance gameplay experience.
- Implemented BFS-based algorithms and game logic to support interactive gameplay, pathfinding, battle system with moving enemies, deterministic regeneration, save/load function, heads up display, and collectible items.

Spam Email Classifier | Python

April 2026

- Built a classification pipeline that processed and vectorized 1,000+ emails using one-hot encoding for spam detection.
- Implemented feature selection and dimensionality reduction to improve model performance and reduce feature redundancy.
- Trained 2 Logistic Regression models with 3 regularization parameters, achieving 91% accuracy.